

HOW MANY FUTURES ARE ENOUGH? LEARNING REDUNDANCY-AWARE EFFECTIVE MULTIMODALITY FOR END-TO-END AUTONOMOUS DRIVING WITH ACTION-CONDITIONED PREDICTIVE REPRESENTATIONS

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ABSTRACT

Multimodal planning has become central to end-to-end autonomous driving because a scene may admit several plausible actions, whereas their conditional average can correspond to none of them. Recent methods have substantially improved the generation and selection of multimodal trajectories, but many practical systems still treat the candidate count as a predefined modeling choice. In reality, the number of decision-relevant alternatives is scene dependent, and multiple trajectories may be geometric variations of nearly interchangeable driving behavior. We study this discrepancy through *effective multimodality*: the scene-dependent capacity needed to preserve distinct planning choices without allocating proposals to redundant outcomes. We introduce a geometry-proposes, predictive-representations-verify framework for estimating this capacity. Trajectory geometry first identifies candidate pairs that are likely redundant. A frozen action-conditioned predictive model then rolls each candidate forward over multiple latent horizons, providing conditional future-consequence evidence for detecting geometrically similar proposals that should not be merged. A lightweight relation verifier converts this evidence into pairwise redundancy decisions that can support subsequent scene-adaptive representative selection. On held-out redundancy evaluation, multi-horizon predictive evidence rejects 17.6% of geometry-induced false merges while retaining 95% of truly redundant pairs, and 27.7% at 90% retention. These results suggest that predictive representations are most useful not as a universal trajectory metric, but as targeted evidence for distinctions that geometry alone cannot resolve.

1 INTRODUCTION

End-to-end autonomous driving must map complex sensory observations and navigation conditions to future ego actions under substantial uncertainty. A single scene can admit several safe and plausible decisions; regressing them to a conditional mean may instead produce an action that matches none of the valid behaviors. Accordingly, multimodality has been modeled through finite trajectory and dense goal sets (Phan-Minh et al., 2020; Gu et al., 2021), probabilistic action vocabularies and multi-target distillation (Chen et al., 2024; Li et al., 2024; 2025a), iterative proposal refinement and selection (Guo et al., 2025; Yao et al., 2025), and diffusion- or flow-based generation (Liao et al., 2025; Xing et al., 2025; Liu et al., 2026). These advances have made planners increasingly capable of producing and ranking high-quality alternatives.

Producing multiple trajectories, however, is not equivalent to representing multiple decisions. Many practical planners instantiate a predefined proposal or sampling budget even though scene complexity varies considerably. An unobstructed road may require only a compact set, while an intersection, merge, or dense interaction can demand more alternatives. Within one scene, several candidates may also be local variants of the same behavior, differing only in offset, timing, or small perturbations. Raw candidate count and geometric spread can therefore overstate decision diversity. A related observation has recently emerged in forecasting: winner-take-all training admits a K-means-like

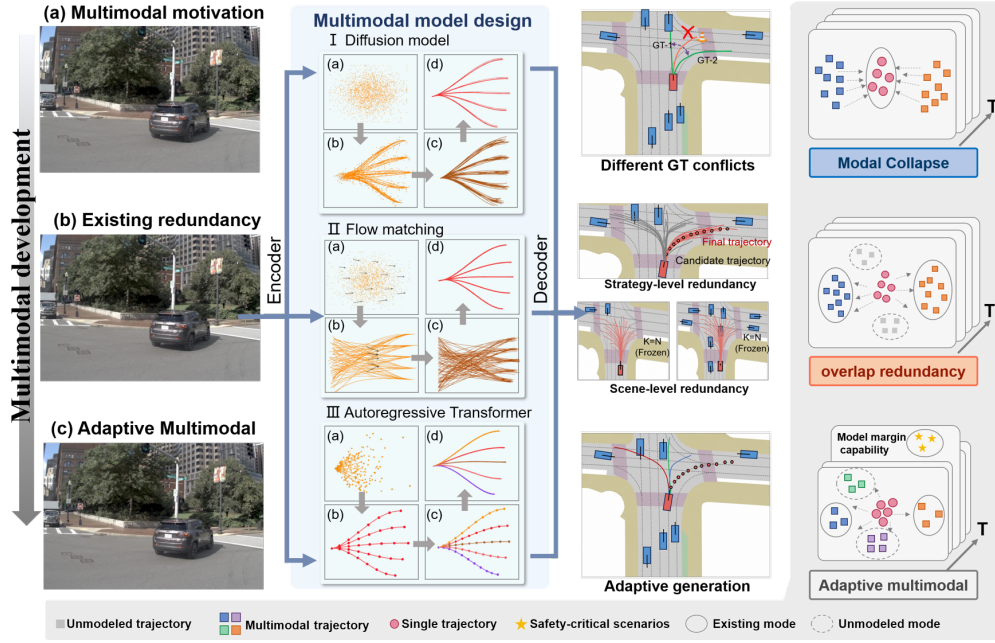


Figure 1: **Motivation for effective multimodality.** Multimodal planners are needed to preserve distinct safe decisions and avoid unsafe averaging, but a fixed proposal budget can contain overlapping policies and allocate unnecessary capacity to simple scenes. We seek a scene-adaptive set that removes consequence-equivalent candidates while retaining safety-critical alternatives.

interpretation that can over-segment trajectory space and assign poorly calibrated distinctions to neighboring modes (Wu et al., 2026). Although this result does not establish redundancy in an end-to-end planner, it independently shows why more output modes need not mean more informative modes.

This motivates a different question: rather than asking how many trajectories a planner can generate, how many decision-relevant futures does a particular scene require? We call this quantity the *effective multimodal capacity*. For a scene s with proposal bank $\mathcal{A}_s$, K_{eff} conceptually denotes the smallest representative subset that preserves the planning utility of the full bank within a tolerance while not increasing safety risk beyond an allowed margin. The objective is therefore not to minimize K at any cost. It is to allocate a compact capacity to simple scenes and retain additional irreducible alternatives when interaction or safety makes them necessary. Figure 1 illustrates this distinction.

Trajectory geometry is the natural first signal for estimating redundancy. Set-based predictors use geometric coverage to span feasible futures efficiently (Phan-Minh et al., 2020; Gu et al., 2021), and trajectory-space distances cheaply expose obvious duplicates. Geometry alone, however, cannot establish consequence equivalence. Spatially close candidates may differ in velocity, interaction timing, clearance, or another agent’s response and thus lead to different safety outcomes. Conversely, visible displacement need not imply a fundamentally different decision. LookOut makes a closely related point from the prediction side by valuing futures that require distinct ego reactions, rather than diversity for its own sake (Cui et al., 2021). Effective capacity consequently requires both an efficient geometric proposal mechanism and evidence about what candidate actions imply.

Predictive representations offer a promising source of such evidence. LAW predicts future latent scene features conditioned on current representations and ego trajectories (Li et al., 2025b), while World4Drive uses intention-aware latent futures to generate and evaluate multimodal plans (Zheng et al., 2025). More broadly, V-JEPA learns visual representations through latent video prediction (Bardes et al., 2024), and V-JEPA 2 extends this line with an action-conditioned variant for planning (Assran et al., 2025). Recent driving models adapt predictive representation learning to multimodal trajectory distillation or explicitly emphasize action-sensitive transitions and joint world-action prediction (Wang et al., 2026a; Zhang et al., 2026; Wang et al., 2026b). Our goal is

not to replace the planner with another world model. Instead, we ask whether a frozen predictive representation can provide conditional evidence about the future consequences of candidate actions.

We address this question with a *geometry proposes, predictive representations verify* framework. A geometric gate first nominates locally similar pairs that are likely redundant. For these ambiguous pairs, a frozen action-conditioned model rolls each candidate forward across multiple latent horizons, and a lightweight relation verifier uses the resulting predictive features to determine whether the proposals can be treated as interchangeable. Verified relations can then support representative selection and a scene-specific K_{eff} . Crucially, we do not globally cluster candidates by latent distance: an action-sensitive representation does not make Euclidean latent distance a valid universal consequence metric. Our held-out analysis reflects this distinction. Global latent distance provides no stable gain in the low-false-merge regime, whereas geometry-gated multi-horizon evidence rejects 17.6% of erroneous geometric merges at 95% retention of truly redundant pairs and 27.7% at 90% retention.

Our contributions are threefold:

- We formulate *effective multimodality* as a scene-conditioned capacity problem, shifting attention from raw proposal count to the number of alternatives required under utility and safety constraints.
- We introduce a consequence-aware consolidation framework in which trajectory geometry efficiently proposes likely redundancy and a frozen action-conditioned predictive representation supplies multi-horizon evidence for verifying geometrically ambiguous pairs.
- We establish a held-out relation evaluation that measures the trade-off between retaining truly redundant pairs and rejecting unsafe geometry-induced merges, and show that conditional predictive evidence recovers distinctions missed by geometry alone.

2 RELATED WORK

2.1 MULTIMODAL TRAJECTORY PREDICTION AND PLANNING

Early multimodal predictors represented uncertainty with finite trajectory or goal sets. CoverNet casts prediction as classification over dynamically feasible trajectories, while DenseTNT starts from dense goals and learns a compact set of futures optimized offline (Phan-Minh et al., 2020; Gu et al., 2021). End-to-end driving has since incorporated richer action distributions. VADv2 learns a probabilistic planning vocabulary, and Hydra-MDP and Hydra-MDP++ distill human and rule-based experts into multi-head planners (Chen et al., 2024; Li et al., 2024; 2025a). Generative planners model multimodality through truncated diffusion, goal-conditioned flow matching, or constraint-guided flow matching (Liao et al., 2025; Xing et al., 2025; Liu et al., 2026). A complementary proposal-centric line iteratively refines candidate plans or improves their coarse-to-fine selection (Guo et al., 2025; Yao et al., 2025); Drive-JEPA further couples proposal-centric planning with multimodal trajectory distillation and predictive video representation learning (Wang et al., 2026a). These methods substantially improve proposal quality, coverage, and selection. Our question is orthogonal to the particular generator: given a strong candidate bank, which proposals still correspond to distinct decisions?

2.2 DIVERSITY, COVERAGE, AND EFFECTIVE MODE CAPACITY

Trajectory-set methods make coverage explicit but optimize criteria different from consequence-preserving consolidation. CoverNet studies fixed and dynamically constructed sets that cover the state space, whereas DenseTNT uses dense goals and offline optimization to provide multi-future supervision for a compact online predictor (Phan-Minh et al., 2020; Gu et al., 2021). LookOut moves beyond geometric variety by constructing futures that elicit different reactions from the ego vehicle (Cui et al., 2021). This reaction-aware view is closely related to our interest in preserving alternatives that matter for planning.

Recent analysis also clarifies why nominal mode count can exaggerate useful diversity. Wu et al. connect winner-take-all forecasting objectives to K-means-like hard assignment and show that they can over-segment trajectory space, destabilize assignments between neighboring modes, and

yield uninformative mode probabilities (Wu et al., 2026). Their posterior-weighted merging and soft-responsibility treatments address calibration and mode pruning in forecasting. We instead consolidate ego-planning proposals, where two candidates are interchangeable only if merging them does not discard a relevant future consequence or safety-critical alternative. Geometry remains a strong and inexpensive proposer for this task, but it does not by itself certify consequence equivalence. Effective multimodality therefore complements work on coverage, calibrated probabilities, and diverse generation with a scene-conditioned capacity criterion.

2.3 PREDICTIVE REPRESENTATIONS AND WORLD MODELS FOR DRIVING

Latent world models use future prediction to improve driving representations and planning. LAW predicts future scene features from the current representation and an ego trajectory, providing self-supervision for end-to-end planning (Li et al., 2025b). World4Drive builds intention-aware physical latent futures for multimodal trajectory generation and evaluation (Zheng et al., 2025). These approaches place learned future representations directly within the driving pipeline.

Joint-embedding predictive architectures provide a broader representation-learning foundation. V-JEPA predicts masked video content in representation space rather than reconstructing pixels (Bardes et al., 2024); V-JEPA 2 scales this approach and includes an action-conditioned latent world model for planning (Assran et al., 2025). Drive-JEPA adapts V-JEPA-style pretraining to end-to-end driving while combining it with multimodal trajectory distillation (Wang et al., 2026a). Delta-JEPA emphasizes action-sensitive latent transitions through latent-difference action decoding, and WA-JEPA jointly models future world representations and ego action for driving (Zhang et al., 2026; Wang et al., 2026b). Rather than using a world model as the planner or treating its latent space as a global clustering metric, we use frozen multi-horizon predictive features as targeted consequence evidence for candidate pairs that geometry cannot confidently resolve.

3 PRELIMINARIES

Problem Definition. Autonomous driving task aims to predict smooth and collision-free trajectory in future seconds, given the ego status S_{ego} (e.g., ego speed and ego acceleration), sensor input I_{cam} , and navigation information L_{nav} . Conventional end-to-end driving algorithm Φ formulate this as

$$\mathbf{V}_{\text{traj}} = \Phi(I_{\text{cam}}, L_{\text{nav}}, S_{\text{ego}}), \quad (1)$$

where $\mathbf{V}_{\text{traj}} \in \mathbb{R}^{T \times 3}$ is the sequence of future waypoints and headings. While methods such as have shown strong effectiveness, their black-box nature impedes interpretability and they often fail to generalize to rare corner cases in real-world driving.

Recent works utilize the rich world knowledge and causal reasoning capabilities of Vision-Language Models for autonomous driving. VLMs output trajectories in textual form and generate explicit reasoning processes:

$$T_{\text{traj}}, T_{\text{reason}} = \text{VLM}(I_{\text{cam}}, L_{\text{nav}}, S_{\text{ego}}). \quad (2)$$

However, we observe an inherent mismatch between the language-formatted trajectory space and the continuous action space, and the autoregressive decoding process can suffer from output collapse, leading to erroneous trajectories.

Diffusion Policy. Denoising Diffusion Probabilistic Models (DDPMs) learn a generative model by reversing a fixed, Markovian noising process that gradually corrupts data with Gaussian noise. Given a clean trajectory $\mathbf{x}_0$, the forward process defines:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \sigma_t^2} \mathbf{x}_{t-1}, \sigma_t^2 \mathbf{I}), \quad t = 1, \dots, T, \quad (3)$$

where σ_t is the noise standard deviation at step t . At inference, trajectories are generated by initializing $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and iteratively denoising:

$$\mathbf{x}_{t-1} = \frac{1}{\sqrt{1 - \sigma_t^2}} \left(\mathbf{x}_t - \sigma_t^2 \epsilon_\theta(\mathbf{x}_t, t) \right) + \sigma_t \mathbf{z}, \quad \mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (4)$$

Denoting trajectory waypoint as $\mathbf{x}_0$, this framework naturally extends to trajectory-level policy generation, where the denoising network ϵ_θ learns to refine noisy motion plans into smooth trajectories.

4 METHODOLOGY

In this section, we present the overall framework and training paradigm of ReCogDrive. First, we introduce a scalable hierarchical data pipeline that curates a multi-level cognitive dataset spanning from perception to reasoning to instill the VLM with nuanced, human-like driving priors. Second, we detail our cognition-guided diffusion planner, a novel architecture that resolves the modality mismatch by translating the VLM’s latent semantic representations into continuous and stable trajectories. Finally, we describe our policy refinement stage, where a Diffusion Group Relative Policy Optimization (DiffGRPO) algorithm is employed to fine-tune the planner with simulated rewards, optimizing for safety and comfort beyond the confines of the expert dataset.

4.1 SCALABLE HIERARCHICAL DATA PIPELINE FOR DRIVING PRETRAINING

To bridge the gap between general Vision-Language Models (VLMs) and autonomous driving, we introduce a scalable, structured pipeline for the automated generation of high-fidelity driving data. This pipeline comprises three stages: Generation, Refinement, and Quality Control.

In the Generation stage, we align with the sequential cognitive process of human drivers, organized into four levels: (1) *Foundational Perception*: constructing representations of static and dynamic scene elements; (2) *Dynamic Understanding*: analyzing multi-agent dynamics and predicting near-term behaviors of surrounding entities; (3) *Planning and Reasoning*: formulating executable, safety-aware driving plans and providing concise rationales; and (4) *Advanced Reasoning*: conducting counterfactual analyses and fine-grained trade-off evaluations. To achieve this, we combine ground-truth data for objective tasks with a state-of-the-art VLM for subjective annotations.

The Refinement stage integrates open-source datasets, followed by normalization, rewriting, and question-template augmentation to ensure linguistic and semantic consistency. Finally, the Quality Control stage applies automated scoring and rigorous filtering based on linguistic accuracy, visual clarity, etc., retaining only the high quality samples. This systematic design ensures our pre-training dataset is both reliable and scalable, enabling the adaptation of a general VLM into a domain-specialized cognitive model for autonomous driving.

4.2 COGNITION-GUIDED DIFFUSION PLANNER

While autoregressive paradigms offer a straightforward way to generate trajectories in textual form, they suffer from inherent limitations: (i) the mismatch between discrete language space and continuous action space often leads to infeasible or jerky trajectories, and (ii) VLMs are prone to hallucination and decoding errors, which compromise safety in autonomous driving. To address these issues, we propose a *Cognition-Guided Diffusion Planner*, which integrates human cognitive driving priors from a VLM with a diffusion planner to generate smooth and safety-aware trajectories. Formally, given a noisy trajectory sample $\mathbf{x}_t \in \mathbb{R}^{N \times 3}$, the denoising step is defined as:

$$\mathbf{x}_{t-1} = D_{\text{act}}\left(\text{DiT}_{\theta}(z_t; F_h; S_{ego}; t)\right), \quad z_t = \text{concat}(E_{\text{act}}(\mathbf{x}_t), E_{\text{his}}(\mathbf{x}_{\text{hist}}), \bar{F}_h), \quad (5)$$

where F_h denotes the VLM hidden states, S_{ego} the ego-vehicle status, and z_t the fusion of noisy actions, historical trajectories, and semantic priors. The diffusion model is trained by minimizing:

$$\mathcal{L}_{\text{dif}} = \mathbb{E}_{z_t, c} \|\epsilon - \epsilon_{\pi}(z_t, c)\|^2, \quad (6)$$

with $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ and condition $c = \{F_h, S_{ego}\}$.

Vision-Language Model Guidance. We adopt InternVL3 as the cognitive backbone for its strong multi-modal reasoning. The VLM encodes multi-view images and textual queries into hidden states $F_h \in \mathbb{R}^{L \times D}$, which act as cognitive tokens carrying driving priors and serve as cross-attention conditions in the diffusion transformer. We further obtain a global semantic embedding $\bar{F}_h$ via average pooling over F_h , providing stable contextual guidance during denoising. Additionally, the VLM can optionally output high-level driving instructions and chain-of-thought reasoning to enhance trajectory generation.

Diffusion Transformer Design. Our diffusion planner builds on DiT blocks, enhanced with lightweight but effective design choices: SwiGLU FFN for expressive non-linearity, RoPE for

relative positional encoding, and both QK-Norm and RMSNorm for stable optimization. Each block alternates between (i) *self-attention*, which models pairwise waypoint relations, and (ii) *cross-attention*, which injects semantic priors F_h into the trajectory space. This design enables a structured fusion of scene-level cognition with low-level trajectory optimization.

Historical and Ego Information Fusion. To capture temporal consistency, historical trajectories $\mathbf{x}_{\text{hist}}$ are encoded and concatenated with the noisy trajectory embeddings. Ego-vehicle status S_{ego} (e.g., speed, acceleration) is injected via *AdaLN modulation*, ensuring that the planner is conditioned on the vehicle’s physical state. This dual conditioning mechanism enhances stability, safety, and adaptability in diverse driving scenarios.

4.3 DIFFUSION GROUP RELATIVE POLICY OPTIMIZATION

Relying solely on imitation learning exhibits fundamental limitations, since expert demonstrations may be multi-modal, leading to conflictual optimization results. When trained in a rare intersection turn scenario with multiple expert trajectories, the model can resort to learning an average trajectory to achieve global optimality, which can lead to incorrect or unsafe maneuvers. Consequently, even though the diffusion planner trained through imitation learning closely matches expert trajectories in terms of displacement, it still frequently produces collisions or drives outside the drivable area.

To enable exploration beyond imitation, we introduce *DiffGRPO*, a Group Relative Policy Optimization algorithm specifically designed for diffusion planners that tightly couples GRPO with the denoising process. Following, we interpret the diffusion policy π_θ as an internal Markov decision process: starting from Gaussian noise, the model gradually denoises to generate a full trajectory. Concretely, we sample G trajectories, each represented by a diffusion chain:

$$\mathbf{x} = (x_T, x_{T-1}, \dots, x_0), \quad (7)$$

where T is the total number of denoising steps and transitions follow:

$$x_T \sim \mathcal{N}(0, \mathbf{I}), \quad x_{t-1} \sim \pi_\theta(x_{t-1} | x_t). \quad (8)$$

Unlike prior approaches that rely on simple ℓ_2 trajectory distances as surrogate rewards, we are the first, to our knowledge, to leverage the NAVSIM simulator to providing realistic feedback on safety and comfort. Each rollout is evaluated in terms of collisions, drivable-area compliance, and driving comfort, with the results aggregated into a Predictive Driver Model Score (PDMS) that serves as the reward r_i . We cast this as a single-step decision, treating the whole trajectory as one composite action with its PDMS score as the reward. We then compute the group-standardized advantage:

$$\hat{A}_i = \frac{r_i - \text{mean}(r_{1..G})}{\sqrt{\text{var}(r_{1..G})}}, \quad i = 1, \dots, G. \quad (9)$$

Each conditional step in the diffusion chain is a Gaussian policy:

$$\pi_\theta(x_{t-1} | x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \sigma_t^2 I), \quad (10)$$

where $\mu_\theta(x_t, t)$ is the model-predicted mean and $\sigma_t^2 I$ the (fixed) covariance. Thus, the probability density of the full chain $\mathbf{x}_{0:T}$ under π_θ , where $p(x_T)$ is a Gaussian prior independent of θ , is:

$$\log \pi_\theta(\mathbf{x}_{0:T}) = \log p(x_T) + \sum_{t=1}^T \log \pi_\theta(x_{t-1} | x_t), \quad (11)$$

Finally, we compute the policy loss following while concurrently incorporating a behavior cloning loss to prevent collapse during exploration.

$$L = \underbrace{-\frac{1}{G} \sum_{i=1}^G \frac{1}{T} \sum_{t=1}^T \gamma^{t-1} \log \pi_\theta(x_{t-1}^{(i)} | x_t^{(i)}) \hat{A}_i}_{L_{\text{RL}}} - \underbrace{\lambda \frac{1}{G} \sum_{i=1}^G \frac{1}{T} \sum_{t=1}^T \log \pi_\theta(\tilde{x}_{t-1}^{(i)} | \tilde{x}_t^{(i)})}_{L_{\text{BC}}}, \quad (12)$$

where γ is the discount coefficient mitigating instability in early denoising steps, λ is the weight for the behavior cloning loss, and $\tilde{x}_{t-1}, \tilde{x}_t$ are values sampled from the reference policy π_{ref} . We omit the PPO-style clipping and set the update iteration to 1, so $\pi = \pi_{\text{old}}$. Through DiffGRPO, the diffusion planner learns to generate safe and comfortable trajectories in closed-loop settings, going beyond mere imitation and enhancing the robustness of our framework.

Table 1: **Performance comparison on NAVSIM *navtest* using closed-loop metrics.** [†] denotes models fine-tuned on the NAVSIM trajectory dataset.

Method	Image	Lidar	NC [†]	DAC [†]	TTC [†]	Comf. [†]	EP [†]	PDMS [†]
Constant Velocity			68.0	57.8	50.0	100	19.4	20.6
Ego Status MLP			93.0	77.3	83.6	100	62.8	65.6
VADv2- $\mathcal{V}_{8192}$	✓		97.2	89.1	91.6	100	76.0	80.9
DrivingGPT	✓		98.9	90.7	94.9	95.6	79.7	82.4
UniAD	✓		97.8	91.9	92.9	100	78.8	83.4
TransFuser	✓	✓	97.7	92.8	92.8	100	79.2	84.0
PARA-Drive	✓		97.9	92.4	93.0	99.8	79.3	84.0
DRAMA	✓	✓	98.0	93.1	94.8	100	80.1	85.5
Hydra-MDP- $\mathcal{V}_{8192}$ -W-EP	✓	✓	98.3	96.0	94.6	100	78.7	86.5
DiffusionDrive	✓	✓	98.2	96.2	94.7	100	82.2	88.1
WoTE	✓	✓	98.5	96.8	94.9	99.9	81.9	88.3
VLMs-based Methods								
QwenVL2.5-8B [†]	✓		97.8	92.1	92.8	100	78.3	83.3
InternVL3-8B [†]	✓		97.0	92.4	91.8	100	78.9	83.3
<i>ReCogDrive(ours)</i>	✓		97.9	97.3	94.9	100	87.3	90.8

Table 2: **Closed-loop and Multi-ability Testing Results in CARLA Bench2Drive Leaderboard.**

Method	Closed-loop Metric [†]				Multi-Ability Test (%) [†]					
	Efficiency	Comfort	Success	DS	Merging	Overtaking	Emerg. Brake	GiveWay	Traf. Sign	Mean
TCP*	54.26	47.80	15.00	40.70	16.18	20.00	20.00	10.00	6.99	14.63
TCP-ctrl*	55.97	51.51	7.27	30.47	10.29	4.44	10.00	10.00	6.45	8.23
TCP-traj*	76.54	18.08	30.00	59.90	8.89	24.29	51.67	40.00	46.28	34.22
TCP-traj w/o distill.	78.78	22.96	30.05	49.30	17.14	6.67	40.00	<u>40.00</u>	28.72	28.51
ThinkTwice	76.93	16.22	3.13	62.44	27.38	18.42	35.82	50.00	54.23	37.17
DriveAdapter*	70.22	16.01	33.08	<u>64.22</u>	<u>28.82</u>	<u>26.38</u>	<u>48.76</u>	50.00	<u>56.43</u>	42.08
AD-MLP	48.45	22.63	0.00	18.05	0.00	0.00	0.00	0.00	4.35	0.87
UniAD-T.	123.92	47.04	13.18	40.73	8.89	9.33	20.00	20.00	15.43	14.73
UniAD-B.	129.21	43.58	16.36	45.81	14.10	17.78	21.67	10.00	14.21	15.55
VAD	157.94	46.01	15.00	42.35	8.11	24.44	18.64	20.00	19.15	18.07
DriveTransformer-L.	100.64	46.01	<u>35.01</u>	63.46	17.57	35.00	48.36	<u>40.00</u>	52.10	38.60
ReCogDrive	<u>138.18</u>	17.45	45.45	71.36	29.73	20.00	69.09	20.00	71.34	<u>42.03</u>

5 EXPERIMENTS

5.1 EXPERIMENTAL SETUP.

Implementation Details. We choose InternVL3 , comprising a 300M-parameter InternViT visual encoder and a Qwen2.5 Large-Language-Model , as our base model, which demonstrates strong performance on multiple benchmarks. Images are processed through a dynamic resolution preprocessing strategy. In the first stage, we conduct supervised fine-tuning (SFT) on the VLM using the dataset constructed by our hierarchical data pipeline for three epochs. In the second stage, with the VLM parameters frozen, we train the diffusion model via DDPM for 200 epochs. In the third stage, we further optimize the diffusion model using reinforcement learning for 10 epochs, with one policy update per batch. Detailed hyperparameter settings are provided in the supplementary material.

Dataset. We evaluate primarily on two challenging benchmarks: NAVSIM and Bench2Drive . NAVSIM is a planning-oriented autonomous driving dataset built on OpenScene , a redistribution of nuPlan . The dataset is split into *navtrain* (1,192 training scenes) and *navtest* (136 evaluation scenes). Bench2Drive is a CARLA-based benchmark composed of 220 short routes, each containing a distinct, safety-critical scenario. In addition, to build the VLM’s cognitive foundation, ReCogDrive is trained on a collection of VQA datasets, including DriveLM and LingoQA , with further details provided in the supplementary material.

Table 3: **Ablation study on the proposed components of the framework.** We evaluate the effect of driving pre-training, diffusion planner, and reinforcement learning on NAVSIM evaluation.

ID	Trajectory training	Driving Pre-training	Diffusion Planner	Diff GRPO	NC	DAC	TTC	Conf.	EP	PDMS
1	✓	✗	✗	✗	97.4	91.3	93.0	100	77.2	82.4
2	✓	✓	✗	✗	97.6	93.1	92.7	100	79.1	84.1
3	✓	✓	✓	✗	98.1	94.7	94.2	100	80.9	86.5
4	✓	✓	✓	✓	97.9	97.3	94.9	100.0	87.3	90.8

Table 4: Comparison of trajectory generation methods.† indicates inference with `lmdeploy`.

Method	Time (s)	PDMS ↑	Err. (%) ↓
VLM Plain Text†	0.5839	84.1	0.01
VLM + MLP	0.0751	75.5	0.00
VLM + Query-based Decoder	0.0750	85.0	0.00
VLM + Vanilla Diffusion	0.0748	85.4	0.00
+ <i>SwiGLU FFN</i>	0.0745	85.7	0.00
+ <i>RMS Norm</i>	0.0749	85.8	0.00
+ <i>RoPE</i> & <i>QK Norm</i>	0.0750	86.5	0.00

Table 5: Model performance on Drive VQA Benchmark.

Method	DriveLM	DriveBench				
	GPT-Score	Percep.	Predict.	Plan.	Behav.	Avg.
LLaVA-1.5	61.91	23.22	22.02	29.15	13.60	22.00
InternVL2	64.13	32.36	45.52	53.27	54.58	46.43
Qwen2-VL	–	30.13	49.35	61.30	51.26	48.01
DriveLM	65.25	16.85	44.33	68.71	42.78	43.17
Dolphins	–	9.59	32.66	52.91	8.81	25.99
GPT-4o	67.27	35.37	51.30	75.75	45.40	51.96
ReCogDrive	67.30	64.95	49.34	70.20	42.36	56.71

5.2 MAIN RESULTS AND ABLATION STUDY

Experiments on the NAVSIM Benchmark. Tab. 1 reports results on NAVSIM. the framework achieves a PDMS of 90.8, establishing a new state-of-the-art. Remarkably, it outperforms Diffusion-Drive and WoTE, both of which use camera and LiDAR inputs, by 2.7 and 2.5 PDMS, respectively, while using only camera input. Compared with our reproduced InternVL3 and QwenVL2.5 trained directly on NAVSIM trajectories, the framework improves PDMS by 7.5, validating the effectiveness of our training paradigm. It also surpasses the prior vision-only state-of-the-art, PARA-Drive, by 6.8 PDMS.

Bench2Drive Closed-loop Performance. Tab. 2 reports closed-loop and multi-ability results on the CARLA Bench2Drive leaderboard. the framework achieves the highest scenario success rate of 45.45% and the top Driving Score of 71.36, surpassing prior end-to-end baselines. It also excels in safety-critical skills such as emergency braking 69.09% and traffic sign compliance 71.34, while maintaining strong efficiency and a competitive multi-ability mean of 42.03. These results highlight the effectiveness and reliability of our framework in complex urban driving.

Ablation study on the framework. Tab. 3 presents an ablation study on the proposed components of the framework. When training InternVL3 solely on NAVSIM trajectory data, the model achieves a PDMS of 83.3. Adapting the VLM to driving scenarios with our large-scale driving QA data increases PDMS by 1.7. Introducing the diffusion planner for continuous trajectory prediction further raises PDMS by 2.4. Finally, DiffGRPO achieves a PDMS of 90.8 with a 4.3 improvement, demonstrating the effectiveness of our RL scheme in producing safer driving behavior.

Comparison of Trajectory Generation Methods. We compare trajectory generation methods in terms of inference speed, PDM Score, and error rate as shown in Tab. 4. Generating trajectories as plain text is slow due to the autoregressive process and may suffer from format errors that compromise safety. In contrast, MLP, Query-based Decoder, and diffusion planner are substantially faster, with diffusion-based approaches further boosting PDMS. Building on this, we integrated lightweight architectural refinements including SwiGLU FFN, RMS Norm, RoPE, and QK Norm. Our final model ReCogDrive achieves a $7.8\times$ speedup and a 2.4 PDMS gain over the text baseline, validating the effectiveness of our design.

Evaluation on Drive VQA Benchmarks. To provide a comprehensive assessment of our model’s language-related capabilities in driving scenarios, we evaluate its performance on the DriveLM and DriveBench benchmarks. The results, presented in Tab. 5, demonstrate the effectiveness of our approach. Our model, ReCogDrive, achieves state-of-the-art performance, surpassing the powerful

closed-source GPT-4o model on most key metrics, including the overall DriveBench average score. Furthermore, ReCogDrive significantly outperforms several specialized open-source driving models, validating its strong visual question answering and reasoning abilities in this domain.

Qualitative Results. We compare ReCogDrive (IL and RL) with Transfuser, where RL yields safer and more reliable trajectories in challenging turning scenarios. More qualitative analysis is provided in the supplementary material.

6 CONCLUSION

In this work, we proposed the framework, a novel reinforced cognitive framework for end-to-end autonomous driving that integrates a Vision-Language Model with a diffusion planner. We first introduced a scalable hierarchical data pipeline that mimics human driving cognition to construct a foundation model with strong driving priors. Building on this foundation, we designed a cognition-guided diffusion planner that injects VLM-derived cognitive tokens into the denoising process, enabling the generation of continuous, stable, and human-like trajectories. To further refine planning behavior, we introduced DiffGRPO, a reinforcement learning scheme that explicitly optimizes safety and comfort in closed-loop driving. Extensive experiments on multiple benchmarks demonstrate that the framework achieves state-of-the-art performance, validating the effectiveness of our approach. We hope our work can advance the development of autonomous driving VLAs.

AI USE STATEMENT

Generative AI tools were used to assist with LaTeX source reconstruction, modularization, formatting, and the drafting of source-code comments and submission materials. All AI-assisted outputs were reviewed by the authors. The authors take responsibility for the final content of the submission, including all text, claims, code, and artifacts produced with the aid of generative AI.

ETHICS STATEMENT

This work studies learning-based autonomous-driving systems, a safety-critical application in which incorrect predictions or plans could cause physical harm if deployed without adequate validation. The experiments described in this manuscript use public research datasets and simulation-based evaluation; the study does not report the collection of new human-subject data. Dataset licenses, privacy constraints, and benchmark terms must be respected when releasing derived annotations and models. The reported system is a research prototype and should not be interpreted as validated for deployment on public roads. Any real-world use would require additional safety engineering, regulatory review, testing under distribution shift, human oversight, and analysis of failure modes and potential demographic or geographic bias.

REPRODUCIBILITY STATEMENT

We provide the model architecture, training objectives, and optimization procedure in the methodology section. Dataset sources and construction steps are documented in Sec. D; evaluation metrics, model hyperparameters, training schedules, software, and hardware configurations are reported in Sec. E; and additional ablations and qualitative examples appear in Secs. C and H. The anonymous supplementary artifact is organized to contain environment specifications, preprocessing instructions, evaluation commands, configuration files, and an expected-results manifest. Random seeds, dataset acquisition instructions, checkpoint provenance, and executable reproduction code must be finalized and verified before submission.

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A APPENDIX

We organize the supplementary material as follows. First, in Sec. B, we address potential questions that may arise from reading the main text. We then report the framework’s performance on the NAVSIM and DriveBench benchmarks, along with more detailed ablation studies in Sec. C. In Sec. D, we provide details of the training data collection and processing pipeline. Sec. E describes the framework’s training and inference implementation, including all key hyperparameters, and also provides details of the evaluation metrics used on NAVSIM and Bench2Drive. Sec. F and Sec. G discuss the limitations of the method and the broader impacts. Finally, Sec. H presents additional qualitative results, including extensive visualizations on NAVSIM and Bench2Drive, comparisons between IL and RL trajectories, driving dialogues, and analyses of failure cases.

B QUESTIONS

Q1. *What is the technical novelty in this paper?*

The key novelty of this work is the first reinforced cognitive framework for autonomous driving that integrates a VLM with a diffusion-based planner. Unlike prior driving VLA approaches (e.g., EMMA , OmniDrive) that generate trajectories purely in the text space, our framework injects VLM-derived driving priors into a diffusion planner, avoiding slow inference, infeasible actions, and trajectory errors. Within this unified framework, we further contribute three innovations: (i) a scalable hierarchical data pipeline that mimics the sequential cognitive process of human drivers through generation, refinement, and quality control, enabling flexible adaptation of VLMs to diverse driving scenarios; (ii) a cognition-guided diffusion planner that leverages VLM-derived priors to efficiently generate continuous and stable trajectories, surpassing plain-text and Transfuser-like decoders; and (iii) Diffusion Group Relative Policy Optimization (DiffGRPO), the first reinforcement learning scheme tailored for diffusion-based planners in autonomous driving VLAs, enabling simulation-driven policy optimization for safer and more comfortable driving.

Q2. *Does the VLM merely act as a feature extractor?*

ReCogDrive consists of a Vision-Language Model and a diffusion planner. In our framework, the VLM can produce high-level instructions or chain-of-thought reasoning to guide planning. However, as shown in Tab. 10, reinforcement learning alone already achieves state-of-the-art performance on the NAVSIM benchmark. Incorporating CoT-based guidance provides no gain, mainly due to the limited diversity of NAVSIM scenarios. Moreover, Bench2Drive already supplies explicit navigation instructions. For efficiency, we therefore keep VLM-based textual guidance optional.

Q3. *Why was the model trained on datasets such as CODA-LM and OmniDrive, yet not evaluated on them?*

Our model is trained on a mixture of 12 open-source driving datasets, which enables handling multiple scenes and tasks. We observed that most existing benchmarks, including CODA-LM and OmniDrive, employ traditional metrics such as BLEU and ROUGE. These metrics evaluate performance by calculating the textual similarity between a model’s response and the ground-truth answer. Such metrics primarily assess dataset-specific fitting rather than general reasoning ability. Therefore, to better evaluate the model’s true Visual Question Answering (VQA) capabilities, we utilize the GPT-Score from the DriveLM and DriveBench benchmarks, as presented in Tab. 5.

Q4. *Does DriveVQA accurately reflect the capabilities of a VLM for autonomous driving?*

DriveBench has shown that many VLMs’ decision accuracy does not degrade with visual quality, indicating a reliance on priors rather than genuine visual understanding. Therefore, we primarily use the Planning task to evaluate a model’s practical capabilities in dynamic driving scenarios. However, for completeness and to facilitate comparison with existing work, we also provide the VQA results on the DriveLM and DriveBench benchmarks.

Q5. *Does ReCogDrive rely on high-quality QA annotations to generalize to other domains?*

ReCogDrive achieves state-of-the-art results by leveraging high-quality driving QA datasets. However, a key contribution of our work is the scalable hierarchical data generation pipeline. Built on advanced VLMs deployed with Sglang, this pipeline can automatically generate large-scale QA annotations, followed by normalization, scoring, and filtering to ensure quality. This design ensures efficiency and adaptability, enabling transfer to new domains without costly manual annotation.

Q6. *Why does applying the formula in Eq. 13 to the averaged sub-metrics in Tab. 1 not yield the reported PDM Score?*

As documented in the NAVSIM issue at <https://github.com/autonomousvision/navsim/issues/116>, the PDMS score is calculated per scene using the formula and is then averaged across all scenes. Applying the formula to sub-metrics that have already been averaged is a different computation and will thus lead to a different final value.

Q7. *Why does the paper adopt such a low discount factor (0.6) for a long-horizon driving task?*

In Eq. 12, the discount factor is applied across denoising steps rather than over the temporal horizon of the driving task. In diffusion-based planning, early denoising steps amplify high-variance noise and often destabilize learning. A smaller discount factor places greater weight on later, more reliable denoising steps, which directly improves stability and feasibility of generated trajectories. This design choice is thus motivated by the dynamics of the denoising process rather than by a short-horizon assumption of the driving task itself.

Q8. *Why does the paper employ a BC loss instead of the standard KL-divergence loss?*

We employ a BC-style objective as a stable surrogate for KL regularization. Direct KL constraints often lead to unstable optimization in diffusion-based RL, since variance amplification across denoising steps can cause noisy and unreliable gradients. In contrast, the BC formulation provides smoother updates and better training stability, which proved crucial for policy learning effectiveness.

Q9. *Does the diffusion planner only provide performance gains?*

As shown in Tab. 4, beyond the +2.4 PDMS improvement over plain-text VLM outputs, the diffusion planner also achieves $7.8\times$ faster inference by requiring only a forward pass rather than autoregressive generation. More importantly, it eliminates the 0.01% template mismatch errors observed in plain-text decoding, an unacceptable rate in safety-critical driving, making it both more accurate and more reliable for practical autonomous driving.

C MORE RESULTS

Table 6: **Performance comparison on Navtest Benchmark with extended metrics.**

Method	NC $\uparrow$	DAC $\uparrow$	EP $\uparrow$	TTC $\uparrow$	C $\uparrow$	TL $\uparrow$	DDC $\uparrow$	LK $\uparrow$	EC $\uparrow$	EPDMS $\uparrow$
Transfuser	97.7	92.8	79.2	92.8	100	99.9	98.3	67.6	95.3	77.8
VADv2	97.3	91.7	77.6	92.7	100	99.9	98.2	66.0	97.4	76.6
Hydra-MDP	97.5	96.3	80.1	93.0	100	99.9	98.3	65.5	97.4	79.8
Hydra-MDP++	97.9	96.5	79.2	93.4	100	100.0	98.9	67.2	97.7	80.6
ARTEMIS	98.3	95.1	81.5	97.4	100	99.8	98.6	96.5	98.3	83.1
ReCogDrive	98.3	95.2	87.1	97.5	98.3	99.8	99.5	96.6	86.5	83.6

Experiments on NAVSIM with extended metrics. Hydra MDP++ introduces additional evaluation metrics: Traffic Light Compliance (TL), Lane Keeping Ability (LK), Driving Direction Compliance (DDC) and Extended Comfort (EC) to more comprehensively assess driving performance. We evaluate the framework on NAVSIM using these extended metrics as well. Tab. 6 compares our approach against existing methods under this metrics. the framework achieves state of the art scores in Driving Direction Compliance (DDC), Lane Keeping Ability (LK), Ego Progress (EP) and Time to

Table 7: **Evaluation of ReCogDrive-VLM on DriveBench.** We report ReCogDrive-VLM’s performance across four key driving dimensions: perception, prediction, planning, and behavior. Results are shown under three input settings: **Clean** (unaltered images), **Corr.** (images with averaged corruptions), and **T.O.** (text-only inputs).

Method	Size	Type	Perception			Prediction			Planning			Behavior		
			Clean	Corr.	T.O.	Clean	Corr.	T.O.	Clean	Corr.	T.O.	Clean	Corr.	T.O.
Human	-	-	47.67	38.32	-	-	-	-	-	-	-	69.51	54.09	-
GPT-4o	-	Commercial	35.37	35.25	36.48	51.30	49.94	49.05	75.75	75.36	73.21	45.40	44.33	50.03
LLaVA-1.5	7 B	Open	23.22	22.95	22.31	22.02	17.54	14.64	29.15	31.51	32.45	13.60	13.62	14.91
LLaVA-1.5	13 B	Open	23.35	23.37	22.37	36.98	37.78	23.98	34.26	34.99	38.85	32.99	32.43	32.79
LLaVA-NeXT	7 B	Open	24.15	19.62	13.86	35.07	35.89	28.36	45.27	44.36	27.58	48.16	39.44	11.92
InternVL2	8 B	Open	32.36	32.68	33.60	45.52	37.93	48.89	53.27	55.25	34.56	54.58	40.78	20.14
Phi-3	4.2 B	Open	22.88	23.93	28.26	40.11	37.27	22.61	60.03	61.31	46.88	45.20	44.57	28.22
Phi-3.5	4.2 B	Open	27.52	27.51	28.26	45.13	38.21	4.92	31.91	28.36	46.30	37.89	49.13	39.16
Oryx	7 B	Open	17.02	15.97	18.47	48.13	46.63	12.77	53.57	55.76	48.26	33.92	33.81	23.94
Qwen2-VL	7 B	Open	28.99	27.85	35.16	37.89	39.55	37.77	57.04	54.78	41.66	49.07	47.68	54.48
Qwen2-VL	72 B	Open	30.13	26.92	17.70	49.35	43.49	5.57	61.30	63.07	53.35	51.26	49.78	39.46
DriveLM	7 B	Specialist	16.85	16.00	8.75	44.33	39.71	4.70	68.71	67.60	65.24	42.78	40.37	27.83
Dolphins	7 B	Specialist	9.59	10.84	11.01	32.66	29.88	39.98	52.91	53.77	60.98	8.81	8.25	11.92
ReCogDrive-VLM	8 B	Specialist	64.95	63.35	71.44	49.34	48.92	53.49	70.20	72.5	71.76	42.36	42.76	44.00

Table 8: Impact of Different BC Loss Weights λ . Table 9: Impact of Different Min Samplings $\sigma_{\min}^{\exp}$.

BC Wt.	NC	DAC	TTC	Conf.	EP	PDMS
0.001	97.9	97.3	94.9	100.0	87.3	90.8
0.01	97.9	97.5	94.9	100.0	86.0	90.3
0.1	97.6	97.2	93.9	100.0	85.0	89.3

Min Samp.	NC	DAC	TTC	Conf.	EP	PDMS
0.01	98.2	97.8	95.6	100.0	82.8	89.5
0.02	97.9	97.3	94.9	100.0	87.3	90.8
0.04	97.9	97.7	95.3	100.0	85.7	90.5

Table 10: Comparison between trajectory-only (Only Traj), adding high-level command (w/ High Com.), and chain-of-thought (w/ CoT).

Mode	NC	DAC	TTC	Conf.	EP	PDMS
Only Traj	97.9	97.3	94.9	100.0	87.3	90.8
w/ High Com.	97.9	97.3	94.9	100.0	87.2	90.8
w/ CoT	97.8	97.2	94.8	100.0	87.1	90.7

Table 11: Impact of different discount factors γ . We use discounting to reduce the influence of early-step noise during the diffusion process.

Discount	NC	DAC	TTC	Conf.	EP	PDMS
0.6	97.9	97.3	94.9	100.0	87.3	90.8
0.8	98.0	97.6	95.3	100.0	85.9	90.5
1.0	97.9	97.5	94.9	100.0	86.0	90.3

Collision (TTC), and delivers a 0.5 improvement in EPDMS over ARTEMIS , demonstrating the effectiveness of our method.

Evaluations of VLMs across driving tasks. Tab. 7 reports results on *DriveBench* across perception, prediction, planning, and behavior. ReCogDrive-VLM achieves the best overall performance and shows strong robustness under both corrupted and text-only settings, outperforming general-purpose VLMs (e.g., Qwen2-VL, InternVL2) and specialist models (e.g., DriveLM, Dolphins). These results demonstrate the strong driving scene understanding capability of our ReCogDrive-VLM.

Impact of Different Behavior Clone Loss Weights. Tab. 8 examines the impact of the BC loss weight λ . As λ decreases, the policy learns more aggressively, and at $\lambda = 0.001$ ego progress increases to 87.3 with slight declines in NC, DAC, and TTC, yielding the highest PDMS.

Impact of Different Min Sampling Values. Following , we apply a cosine schedule to the diffusion noise variances σ_k and clip them to a minimum value $\sigma_{\min}^{\exp}$. Clipping σ_k to a nonzero minimum encourages the diffusion policy to sample more diverse trajectories, yet overly large $\sigma_{\min}^{\exp}$ can destabilize training. Setting $\sigma_{\min}^{\exp} = 0.02$ achieves a PDMS of 90.8, as shown in Tab. 9.

Effect of VLM guidance modes. As shown in Tab. 10, outputting only trajectory achieves performance comparable to adding high-level command guidance, with almost identical PDMS scores. Interestingly, incorporating chain-of-thought reasoning does not further improve the results; instead, it slightly decreases the PDMS score by 0.1. This suggests that the current NAVSIM benchmark lacks sufficient diversity to demonstrate the benefits of high-level command and CoT reasoning, so we keep these guidance optional.

Impact of Different Discount Factors. Tab. 11 shows the impact of the discount factor γ on RL fine-tuning. When $\gamma = 1.0$, all timesteps, including the very noisy early steps, contribute equally to the policy gradient, which amplifies high variability noise and destabilizes learning. In contrast, setting $\gamma = 0.6$ focuses the update on later, more reliable steps and achieves the best PDMS of 90.8.

D TRAINING DATASETS CONSTRUCTION

D.1 DATA COLLECTION

We collect 12 open-source driving QA datasets, including Talk2Car , SUTD , NuScenes-QA , OmniDrive , and others, yielding over 3.1 million question-answer pairs that cover perception, prediction, and planning tasks across diverse real-world scenarios.

Talk2car is built on the nuScenes dataset and contains 850 videos from the nuScenes training set. This dataset has 11,959 natural language commands.

SUTD contains 10,080 in-the-wild videos and annotated 62,535 QA pairs. These videos are obtained through a combination of online collection and offline shooting, covering various weather conditions, times, and road conditions.

DRAMA is a dataset collected to investigate risk location and natural language description in driving scenarios. It contains 17,785 interactive driving scenarios.

NuScenes-QA encompasses 34,149 complex autonomous driving scenes and 459,941 question-answer pairs, with various types of questions. It aims to evaluate a model’s ability to understand and reason about complex visual data in multi-modal, multi-frame, and outdoor scenarios.

DriveGPT4 is built based on the BDD-X dataset, which contains about 20,000 samples. By dividing them into 16,803 training segments and 2,123 testing segments, question-answer pairs are generated using the control signal data and text annotations.

LingoQA contains 28K unique short video scenarios and 419K annotations. The dataset covers various questions related to driving scenarios, including aspects such as behaviors, scenery, and perception.

DriveLM consists of a training set of 4,072 frames and a validation set of 799 frames, with an average of 91.4 QA pairs per frame.

MAPLM contains point-cloud BEV projections and surround view images of various traffic scenes, such as highways and urban roads, and is equipped with element-level, lane-level, and road-level scene descriptions.

NuInstruct is a dataset constructed based on Nuscenes , containing 91K multi-view video instruction-response pairs in 17 subtasks.

CODA-LM comprises 9,768 real-world driving scenarios with 41,722 textual annotations for critical road entities and 21,537 annotations for road corner cases.

OminiDrive covers 3D perception, reasoning, and planning tasks, including offline and online question-response tasks.

Senna design a series of planning-oriented QAs including scene description, traffic signal detection, vulnerable road user identification, motion intention prediction, meta-action planning and planning explanation. Since the senna dataset is not publicly available, we used Qwen2.5-VL-72B for question-answer data annotation.

D.2 SCALABLE HIERARCHICAL DATA PIPELINE

To systematically enhance VLM performance for autonomous driving, we design a scalable hierarchical data pipeline consisting of three stages: *Generation*, *Refinement*, and *Quality Control*. This design allows us to construct diverse, semantically consistent, and high-quality annotations at scale.

Generation. We first leverage Qwen2.5-VL with crafted prompts to generate annotations across the full spectrum of autonomous driving tasks on NAVSIM . These tasks span perception (e.g.,

scene description, key object identification, road marking recognition, traffic light classification, vulnerable road user detection), prediction (e.g., motion prediction), planning (e.g., high-level command prediction and decision explanation), and advanced reasoning (e.g., counterfactual scenarios inspired by OmniDrive). To strengthen supervision, we further incorporate NAVSIM’s existing sensor and trajectory labels into the generation process. Applying this stage yields a large pool of candidate QA pairs.

Refinement. Since open-source datasets employ heterogeneous formats, we unify them through normalization and linguistic enrichment. (i) *Data Normalization:* Different datasets adopt different bounding-box styles. For example, DriveLM represents an object as “<c2,CAM_BACK,864.2,468.3>”, while NuInstruct uses “<car>[c2,584,478,603,516]”. We convert all such variants into a standardized tag format: “<car><FRONT_VIEW><box>[x_1, y_1, x_2, y_2]</box>” following the InternVL3 pre-training convention. Coordinates x_1, y_1, x_2, y_2 are rescaled to the integer range [0,1000] relative to image resolution. (ii) *Data Augmentation:* To enrich linguistic diversity, especially for datasets with limited QA templates (e.g., CODA-LM with only three formats, DRAMA with terse answers), we employ a two-step process. First, an LLM paraphrases question templates to expand structural variation. Second, Qwen2.5-VL is prompted with the original image, question, and answer to generate more detailed, natural responses. This refinement ensures consistency in structure and greater variety in language.

Quality Control. Finally, we implement a rigorous filtering step. Qwen2.5-VL assigns each QA pair a quality score based on predefined linguistic and semantic criteria. Any pair scoring below 60 is discarded. After this filtering process, we retain 2.3M high-quality QA pairs. Notably, applying the pipeline to NAVSIM produces 775K well-structured pairs, which we use to fine-tune the VLM, substantially enhancing its planning and reasoning capabilities.

D.3 DATASET STATISTICS

The dataset consists of 41% perception samples, 11% prediction samples, and 24% planning samples, aimed at improving VLMs’ understanding of driving scenarios. We also include 24% visual instruction samples to maintain the model’s instruction-following ability.

E EXPERIMENT DETAILS

In this section, we first introduce the evaluation metrics of NAVSIM and Bench2Drive, and then detail our model configuration, the hyperparameters used in the three-stage training and evaluation, and our hardware setup.

Evaluation Metrics. We evaluate our method on two primary benchmarks. For the planning-oriented NAVSIM benchmark, we use the official Predictive Driver Model Score (PDMS), a pseudo-simulation-based metric that holistically assesses safety (NC, DAC, TTC), comfort (C), and progress (EP):

$$\text{PDMS} = \text{NC} \times \text{DAC} \times \left(\frac{5 \cdot \text{EP} + 5 \cdot \text{TTC} + 2 \cdot \text{C}}{12} \right). \quad (13)$$

NAVSIM protocol and version scope. NAVSIM is derived from OpenScene and nuPlan and evaluates predicted trajectories through non-reactive pseudo-simulation. Under the NAVSIM v1 *navtest* protocol used by the results in this paper, the rule-based subscores are No at-fault Collision (NC), Drivable Area Compliance (DAC), Time to Collision within bounds (TTC), Ego Progress (EP), and Comfort (C), which are aggregated into PDMS as defined in Eq. 13. This simulation-based scoring is intended to assess safety, progress, and driving quality while retaining an open-loop evaluation setup. Unless explicitly stated otherwise, the NAVSIM results reported in the main paper use this PDMS protocol rather than the NAVSIM v2 EPDMS protocol.

NAVSIM v2 and Extended PDMS. NAVSIM v2 strengthens driving-quality evaluation by extending PDMS with Driving Direction Compliance (DDC), Traffic Light Compliance (TLC), Lane Keeping (LK), History Comfort (HC), and Extended Comfort (EC). The resulting Extended

Predictive Driver Model Score (EPDMS) is

$$\text{EPDMS} = \text{NC} \times \text{DAC} \times \text{DDC} \times \text{TLC} \times \frac{5(\text{EP} + \text{TTC}) + 2(\text{LK} + \text{HC} + \text{EC})}{16}. \quad (14)$$

Driving Direction Compliance (DDC). DDC checks whether the ego vehicle follows the legal direction of travel within lanes and avoids driving in oncoming lanes outside intersections.

Traffic Light Compliance (TLC). TLC evaluates whether the ego vehicle obeys traffic-light phases and enters an intersection only under a valid green signal.

Lane Keeping (LK). LK measures whether the ego vehicle remains near the centerline of its current lane and avoids lingering between adjacent lanes. The metric is disabled at intersections, where reference centerlines may not match the lane markings perceived by the sensors.

History Comfort (HC). HC evaluates continuity between the predicted trajectory and the recent motion history. The predicted trajectory is prepended with a short segment of the human driver’s recent motion using a fixed padding length of 1.5 seconds, and the resulting continuous trajectory is assessed with the comfort metric used by the nuPlan framework.

Extended Comfort (EC). EC checks whether predicted motion remains smooth across consecutive planning time steps, including consistency of the resulting dynamic states.

For the CARLA-based Bench2Drive benchmark, which focuses on safety-critical scenarios, we report the Success Rate (SR) and the average Driving Score (DS). The DS combines Route Completion (RC) with multiplicative penalties for infractions (IS):

$$\text{SR} = \frac{N_{\text{success}}}{N_{\text{total}}}; \quad \text{DS} = \frac{1}{N_{\text{total}}} \sum_{i=1}^{N_{\text{total}}} \left(\text{RC}_i \times \prod_j \text{IS}_{i,j} \right). \quad (15)$$

Model Architecture and Hyperparameter Details. Our model architecture consists of two main components: Vision-Language Models (VLMs) and a diffusion-based trajectory planner. For the Vision-Language Models, we utilize the pre-trained InternVL3-8B and InternVL3-2B model. The model processes visual inputs by splitting the image into patches, with a maximum of 12 dynamic patches allowed. The input images are cropped into 448x448 pixel patches. For the diffusion planner, we use the DiT architecture, employing DDPM/DDIM denoising methods. During training, the denoising process involves 100 steps, while for inference, the diffusion process is reduced to 5 steps. The hidden layer size of the DiT architecture is set to 384, with a head size of 8 and 16 layers.

Training Configuration. In the first stage, we fine-tune the VLM on the combined driving QA dataset for three epochs, with all parameters unfrozen and a batch size of 1,024. We use AdamW with a base learning rate of $4e^{-5}$, weight decay of 0.05, and a cosine learning rate schedule with a 10% linear warmup. In the second stage, we train the diffusion-based planner with behavior cloning for 200 epochs and a batch size of 512. We use AdamW with a learning rate that warms up to $1e^{-4}$ over the first 1.5% of steps, then decays cosinly to $1e^{-6}$, and a weight decay of $1e^{-4}$. In the third stage, we perform simulator-assisted reinforcement learning for 10 epochs with a batch size of 128, using AdamW and a cosine schedule that decays the learning rate from $1e^{-4}$ to 0. Additionally, we stabilize diffusion training and inference by clipping the reconstructed clean estimate x_0 to ± 1.0 , clipping Gaussian noise samples to ± 5.0 , enforcing a nonzero floor on the denoising standard deviation σ_t of 0.02 for training, clamping σ_t to at least 0.10 when computing $\log p(x_{t-1} | x_t)$, and applying a discount factor $\gamma = 0.6$ during RL fine-tuning to downweight early, noisy timesteps.

Hardware Configuration. We implement the framework on Debian with PyTorch, training across four nodes—each equipped with an Intel® Xeon® Platinum 8457C CPU and eight NVIDIA H20 GPUs (32 GPUs in total). Inference is performed on a single node with 8 GPUs.

F LIMITATIONS

Although the framework achieves state of the art performance on the NAVSIM benchmark, it still faces several limitations. These include the difficulty of processing multiple camera inputs, handling

Table 12: **Hyperparameters for the framework.**

Stage	Parameter	Value
I	Number of epochs	3
	Batch size	1024
	Learning rate	$4e^{-5}$
	Weight decay	0.05
	Warmup ratio	0.10
	Learning rate schedule	Cosine
II	Number of epochs	200
	Batch size	512
	Learning rate	$1e^{-4}$
	Learning rate schedule	Cosine
	Weight decay	1×10^{-4}
III	Number of epochs	10
	Batch size	128
	Learning rate	$1e^{-4}$
	Learning rate schedule	Cosine
	BC loss weight	$1e^{-2}$
	Denoised clip threshold	± 1.0
	Noise clip threshold	± 3.0
	Minimum denoising std	0.02
	Minimum log-variance std	0.10
	Discount factor	0.6

video frame sequences, and relatively high inference latency. Future work may address these issues by designing a 3D vision encoder that aligns with textual features and developing more efficient model architectures. We also plan to deploy and evaluate our model on real vehicles.

G BROADER IMPACTS

Our research promotes the application of Vision-Language Models (VLMs) in the autonomous driving domain. Through simulator-assisted reinforcement learning, our model more effectively mitigates collision risk and generalizes to rare, challenging scenarios. This advancement could lead to safer, more reliable autonomous systems capable of handling real-world driving scenarios.

H QUALITATIVE RESULTS

We first discuss qualitative behavior of the framework on NAVSIM and Bench2Drive, focusing on safe and smooth trajectories. Next, we compare ReCogDrive-IL and ReCogDrive-RL to describe the effect of the proposed DiffGRPO. We further discuss dialogue examples generated by ReCogDrive and analyze representative failure cases on NAVSIM to provide insights into current limitations.

H.1 QUALITATIVE RESULTS ON NAVTEST SPLITS

Representative cases from the Navtest splits highlight the framework’s ability to follow navigation instructions while ensuring safety and smoothness.

H.2 QUALITATIVE RESULTS ON BENCH2DRIVE TEST SCENARIOS

We evaluate the framework on challenging scenarios from Bench2Drive, demonstrating its robustness under diverse traffic conditions and rare corner cases, while consistently maintaining strong performance in closed-loop settings.

H.3 COMPARISON BEFORE AND AFTER REINFORCEMENT LEARNING

We contrast Transfuser, imitation learning (IL), and reinforcement learning (RL) variants of the framework, demonstrating the progressive improvements in decision making, trajectory stability, and safety awareness.

H.4 FAILURE CASES

Representative failure cases of the framework on the Navsim benchmark include instances of aggressive driving caused by the model’s tendency to prioritize ego progress, leading to assertive maneuvers. In addition, the relatively weak perception capability of current VLMs means that directly feeding multi-view camera inputs sometimes results in suboptimal performance, particularly in turning scenarios. We also observe unsafe following distances due to insufficient modeling of surrounding vehicle behaviors, reflecting the limitations of VLMs in motion prediction. Finally, we identify cases where predicted trajectories are visually close to the ground truth but still scored as failures, suggesting that the evaluation metrics may suffer from false negatives.

H.5 MODEL–HUMAN DIALOGUE

The dialogue analysis considers detailed scene descriptions, traffic-sign recognition, reasoning about driving intent, and safe turning-trajectory generation.